28 Which statement best describes the 'electron cloud' in an atom?

A A shell of negative charge at a fixed radius.

B A region where the electron's charge is uniformly distributed.

C The path the electron follows around the nucleus.

D A region where the probability of finding the electron is high.

Ans: **D**

The state of the electron of the atom is represented as a wavefunction ψ and the square of the wavefunction amplitude $|\psi|^2$ is the probability density function.

An electron cloud is the visual representation of the probability density function which dictates where an electron is likely to be found around the nucleus of the atom.